MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 9

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Recap of Previous Lecture

ELBO for Gaussian Diffusion Model

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \log p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1) - D_{\mathrm{KL}}\big(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T)\big) - \sum_{t=2}^{T} \underbrace{\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\mathrm{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\|p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big)}_{\mathcal{L}_t}$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t,\mathbf{x}_0),\tilde{\beta}_t\mathbf{I}),$$
$$p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\big(\mathbf{x}_{t-1}|\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t),\boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t)\big)$$

It is assumed that $\boldsymbol{\sigma}^2_{\boldsymbol{\theta},t}(\mathbf{x}_t) = \tilde{\beta}_t\mathbf{I}$.

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left[\frac{1}{2\tilde{\beta}_t}\left\|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t,\mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t)\right\|^2\right]$$

Recap of Previous Lecture

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[\frac{1}{2\tilde{\beta}_t} \left\| \tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) \right\|^2 \right]$$

Reparametrization

$$\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}$$

$$\boldsymbol{\mu}_{\boldsymbol{\theta},t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t)$$

Recap of Previous Lecture

$$\mathcal{L}_t = \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left[\frac{(1-\alpha_t)^2}{2\tilde{\beta}_t\alpha_t(1-\bar{\alpha}_t)}\left\|\boldsymbol{\epsilon}-\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0+\sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}\right)\right\|^2\right]$$

At every step of the reverse process, we attempt to predict the noise $\boldsymbol{\epsilon}$ that was used in the forward diffusion process!

Simplified Objective

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})}\left\|\boldsymbol{\epsilon}-\boldsymbol{\epsilon}_{\boldsymbol{\theta},t}\left(\sqrt{\bar{\alpha}_t}\cdot\mathbf{x}_0+\sqrt{1-\bar{\alpha}_t}\cdot\boldsymbol{\epsilon}\right)\right\|^2$$

Recap of Previous Lecture

Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}$.
3. Compute loss $\mathcal{L}_{\text{simple}} = \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta}, t}(\mathbf{x}_t)\|^2$.

Sampling (Ancestral)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute the mean of $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta}, t}(\mathbf{x}_t), \tilde{\beta}_t \cdot \mathbf{I})$:

$$\boldsymbol{\mu}_{\boldsymbol{\theta}, t}(\mathbf{x}_t) = \frac{1}{\sqrt{\alpha_t}} \cdot \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t(1 - \bar{\alpha}_t)}} \cdot \boldsymbol{\epsilon}_{\boldsymbol{\theta}, t}(\mathbf{x}_t).$$

3. Denoise $\mathbf{x}_{t-1} = \boldsymbol{\mu}_{\boldsymbol{\theta}, t}(\mathbf{x}_t) + \sqrt{\tilde{\beta}_t} \cdot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.

Recap of Previous Lecture

DDPM Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)}\mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left[C_{2,t}\left\|\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0)\right\|_2^2\right]$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\cdot\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\cdot\boldsymbol{\epsilon}$$

In practice, this coefficient is often omitted.

NCSN Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)}\mathbb{E}_{t\sim U\{1,T\}}\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t|\mathbf{x}_0)\right\|_2^2$$

Note: The objectives of DDPM and NCSN are almost identical; however, their sampling procedures differ:

- NCSN utilizes annealed Langevin dynamics,
- DDPM employs ancestral sampling.

Recap of Previous Lecture

Summary

- Gaussian perturbations:
  - DDPM: $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \cdot \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \cdot \boldsymbol{\epsilon}$;
  - NCSN: $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \cdot \boldsymbol{\epsilon}$.
  - Rescaling the DDPM state by $1/\sqrt{\bar{\alpha}_t}$ gives the NCSN noise form.
- The objectives coincide: ELBO $\equiv$ score-matching.
- The sampling procedures differ:
  - Ancestral sampling in DDPM;
  - Annealed Langevin dynamics for NCSN;
  - Hybrid approaches that combine both updates are possible.

Kingma D. et al. Variational Diffusion Models, 2021
Song Y. et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Recap of Previous Lecture

Sampling (DDIM)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$; choose a grid $T = t_K > \cdots > t_0 = 0$.
2. For $k = K, \ldots, 1$, set $t = t_k$, $s = t_{k-1}$ and compute:

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_{\boldsymbol{\theta},t}(\mathbf{x}_t) \qquad \text{predict noise,}$$

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\,\hat{\boldsymbol{\epsilon}}}{\sqrt{\bar{\alpha}_t}} \qquad \text{estimate clean data,}$$

$$\mathbf{x}_s = \sqrt{\bar{\alpha}_s}\,\hat{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_s}\,\hat{\boldsymbol{\epsilon}} \qquad \text{update state.}$$

3. Return $\mathbf{x}_0$.

Deterministic limit: fresh transition noise is zero ($\sigma_{t\to s} = 0$); randomness comes from $\mathbf{x}_T$.

Fewer steps: choose $K \ll T$; K steps require K network evaluations and no retraining.

Recap of Previous Lecture

DDPM Sampling

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \sigma_t \cdot \boldsymbol{\epsilon}$$

Guided Generation

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) + \sigma_t \cdot \boldsymbol{\epsilon}$$

Guided Generation

$$\begin{aligned}\nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) &= \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) \\ &= \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)\end{aligned}$$

Here, $p(\mathbf{y}|\mathbf{x}_t)$ denotes a classifier operating on noisy samples (which must be trained separately).

Dhariwal P., Nichol A. Diffusion Models Beat GANs on Image Synthesis, 2021

Recap of Previous Lecture

Guided Score Function

$$\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$
$$\mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) = \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$$

Note: Increasing γ sharpens $p(\mathbf{y}|\mathbf{x}_t)$, increasing the contrast

$$\hat{p}(\mathbf{y}|\mathbf{x}_t) \propto p(\mathbf{y}|\mathbf{x}_t)^{\gamma}.$$

Very large γ can reduce diversity and introduce artifacts.

Guidance Interval

Apply extra guidance only at intermediate noise levels. Outside the interval, $\gamma_t = 1$: **conditioning remains**.

Dhariwal P., Nichol A. Diffusion Models Beat GANs on Image Synthesis, 2021
Kynkäänniemi T. et al. Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models, 2024 · Sadat S. et al. Eliminating Oversaturation and Artifacts of High Guidance Scales in Diffusion Models, 2025

Recap of Previous Lecture

Training

1. Train the DDPM as before.
2. Train an additional classifier $p(\mathbf{y}|\mathbf{x}_t)$ on noisy data (time-dependent).

Sampling (Guided)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
2. Denoise with the scaled guided score function:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \cdot \mathbf{x}_t + \frac{\beta_t}{\sqrt{1-\beta_t}} \cdot \mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) + \sigma_t \cdot \boldsymbol{\epsilon}.$$

Recap of Previous Lecture

The previous method requires an additional classifier $p(\mathbf{y}|\mathbf{x}_t)$ trained on noisy data. Let's try to avoid this requirement.

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) = \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t)$$

$$\begin{aligned}
\nabla^{\gamma}_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) &= \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t) \\
&= \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \gamma \cdot \big(\nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y}) - \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t)\big) \\
&= (1-\gamma) \cdot \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t) + \gamma \cdot \nabla_{\mathbf{x}_t} \log p_{\boldsymbol{\theta}}(\mathbf{x}_t|\mathbf{y})
\end{aligned}$$

Ho J., Salimans T. Classifier-Free Diffusion Guidance, 2022
Holderrieth P., Erives E. An Introduction to Flow Matching and Diffusion Models, 2025

Recap of Previous Lecture

Scaled Guided Score Function

$$\mathbf{s}^{\gamma}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}) = (1-\gamma)\cdot \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) + \gamma \cdot \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$$

CFG

1. Introduce the "absence of conditioning" label $\mathbf{y} = \varnothing$.
2. Identify the unguided score function $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) = \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing)$.
3. Train a single model $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$ on **supervised** data, dropping the label $\mathbf{y}$ with some fixed probability (simulating $\mathbf{y} = \varnothing$).
4. At inference, evaluate the model twice to obtain $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing)$ and $\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y})$.

Guidance distillation: train a student to match the fixed teacher's guided prediction:

$$\mathbf{s}_{\boldsymbol{\phi},t}(\mathbf{x}_t, \mathbf{y}, \gamma) \approx (1-\gamma)\mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \varnothing) + \gamma \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t, \mathbf{y}).$$

Evaluate the student **once per step**; keep the sampler and step count.

Ho J., Salimans T. Classifier-Free Diffusion Guidance, 2022
Holderrieth P., Erives E. An Introduction to Flow Matching and Diffusion Models, 2025
Meng C. et al. On Distillation of Guided Diffusion Models, 2023

Outline

01 Continuous-Time Normalizing Flows (CNF)

02 Continuity Equation for CNF Log-Likelihood

03 SDE Basics

Outline

01 **Continuous-Time Normalizing Flows (CNF)**

02 Continuity Equation for CNF Log-Likelihood

03 SDE Basics

Discrete-Time Normalizing Flows

Change of Variable Theorem (CoV)

Let $\mathbf{x}$ be a random variable with density $p(\mathbf{x})$, and let $\mathbf{f} : \mathbb{R}^m \to \mathbb{R}^m$ be a differentiable and **invertible** transformation. If $\mathbf{z} = \mathbf{f}(\mathbf{x})$, $\mathbf{x} = \mathbf{f}^{-1}(\mathbf{z}) = \mathbf{g}(\mathbf{z})$, then

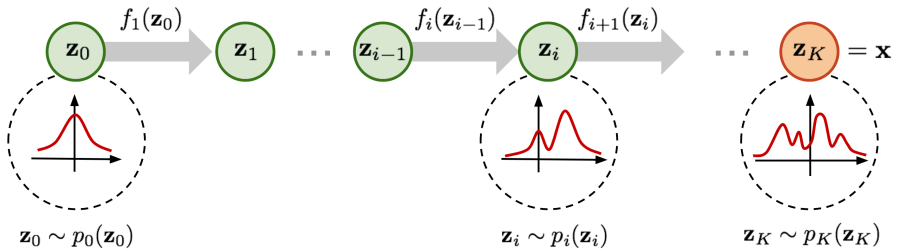
$$
\begin{aligned}
p(\mathbf{x}) &= p(\mathbf{z}) |\det(\mathbf{J}_\mathbf{f})| = p(\mathbf{z}) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right| \\
&= p(\mathbf{f}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} \right) \right|
\end{aligned}
$$

Discrete-Time Normalizing Flows

Change of Variable Theorem (CoV)

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$$
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&= p(\mathbf{f}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} \right) \right|
\end{aligned}
$$



$$
\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \log p(\mathbf{f}_K \circ \cdots \circ \mathbf{f}_1(\mathbf{x})) + \sum_{k=1}^{K} \log \left| \det \left(\frac{\partial \mathbf{f}_k}{\partial \mathbf{f}_{k-1}} \right) \right| .
$$

Towards Continuous-Time Normalizing Flows

Up to this point, we have considered discrete-time normalizing flows:

$$\mathbf{x}_{t+1} = \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}_t, t); \quad \log p(\mathbf{x}_{t+1}) = \log p(\mathbf{x}_t) - \log \left| \det \frac{\partial \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}_t)}{\partial \mathbf{x}_t} \right|.$$

Behrmann J. et al. Invertible Residual Networks, 2019
Chen R. T. Q. et al. Residual Flows for Invertible Generative Modeling, 2019

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Residual Flows

Here $\mathbf{f}_{\boldsymbol{\theta}}$ is the NF **bijection**, $\mathbf{v}_{\boldsymbol{\theta}}$ is the **velocity** (vector field):

$$\mathbf{x}_{t+1} = \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) = \mathbf{x}_t + \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$$

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Here $\mathbf{f}_{\boldsymbol{\theta}}$ is the NF **bijection**, $\mathbf{v}_{\boldsymbol{\theta}}$ is the **velocity** (vector field):

$$\mathbf{x}_{t+1} = \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) = \mathbf{x}_t + \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$$

Given $\mathbf{y} = \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}, t)$, recover $\mathbf{x}$ by **fixed-point iteration**:

$$\mathbf{x} = \mathbf{y} - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t), \qquad \mathbf{x}^{(k+1)} = \mathbf{y} - \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}^{(k)}, t).$$

If $\mathbf{v}_{\boldsymbol{\theta}}$ has Lipschitz constant $L < 1$ in $\mathbf{x}$, the iteration is contractive. **Banach's theorem** guarantees a unique solution for every $\mathbf{y}$, so $\mathbf{f}_{\boldsymbol{\theta}}$ is invertible.

Behrmann J. et al. Invertible Residual Networks, 2019
Chen R. T. Q. et al. Residual Flows for Invertible Generative Modeling, 2019

Towards Continuous-Time Normalizing Flows

A residual step $\mathbf{x}(t+h) = \mathbf{x}(t) + h\,\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$ is invertible if $hL < 1$:

$$\frac{\mathbf{x}(t+h) - \mathbf{x}(t)}{h} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

Towards Continuous-Time Normalizing Flows

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$$\frac{\mathbf{x}(t+h) - \mathbf{x}(t)}{h} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

Taking the limit $h \to 0$, we obtain continuous-time dynamics.

Continuous-Time Dynamics

Consider an Ordinary Differential Equation (ODE):

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0.$$

Towards Continuous-Time Normalizing Flows

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$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0.$$

$$\mathbf{x}(t_1) = \int_{t_0}^{t_1} \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)\,dt + \mathbf{x}_0$$

The **velocity field** $\mathbf{v}_{\boldsymbol{\theta}} : \mathbb{R}^m \times [t_0, t_1] \to \mathbb{R}^m$ need not be contractive or invertible. Under suitable regularity conditions, **uniqueness of ODE solutions guarantees an invertible flow**.

ODEs, Flows and Numerical Solutions

Ordinary Differential Equation

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t), \qquad \mathbf{x}(t_0) = \mathbf{x}_0.$$

ODEs, Flows and Numerical Solutions

Ordinary Differential Equation

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t), \qquad \mathbf{x}(t_0) = \mathbf{x}_0.$$

Flow

The **flow** $\boldsymbol{\psi} : \mathbb{R}^m \times [t_0, t_1] \to \mathbb{R}^m$ maps the initial point to its position at time t: $\mathbf{x}(t) = \boldsymbol{\psi}_t(\mathbf{x}_0)$.

$$\frac{d\boldsymbol{\psi}_t(\mathbf{x}_0)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\boldsymbol{\psi}_t(\mathbf{x}_0), t), \qquad \boldsymbol{\psi}_{t_0}(\mathbf{x}_0) = \mathbf{x}_0.$$

ODEs, Flows and Numerical Solutions

Ordinary Differential Equation

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t), \qquad \mathbf{x}(t_0) = \mathbf{x}_0.$$

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$$\frac{d\boldsymbol{\psi}_t(\mathbf{x}_0)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\boldsymbol{\psi}_t(\mathbf{x}_0), t), \qquad \boldsymbol{\psi}_{t_0}(\mathbf{x}_0) = \mathbf{x}_0.$$

Numerical Solution

$$\boldsymbol{\psi}_t(\mathbf{x}_0) = \mathbf{x}_0 + \int_{t_0}^{t} \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(s), s)\, ds \approx \texttt{ODESolve}_v(\mathbf{x}_0, \boldsymbol{\theta}, t_0, t).$$

A numerical solver approximates the flow by a sequence of **finite steps**.

Numerical Solution of ODEs: Euler

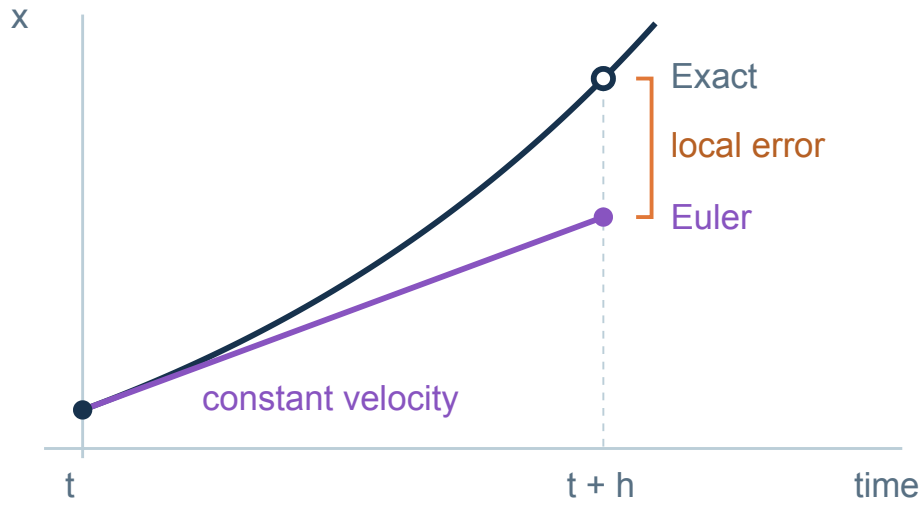
Euler Update Step

$$\frac{\mathbf{x}(t+h) - \mathbf{x}(t)}{h} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

$$\mathbf{x}(t+h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

Freeze the current velocity over the step: **one field evaluation**.

One step: follow the tangent



Euler method; original one-step illustration
Driscoll T. A., Braun R. J. Fundamentals of Numerical Computation: Runge–Kutta methods

Numerical Solution of ODEs: Euler

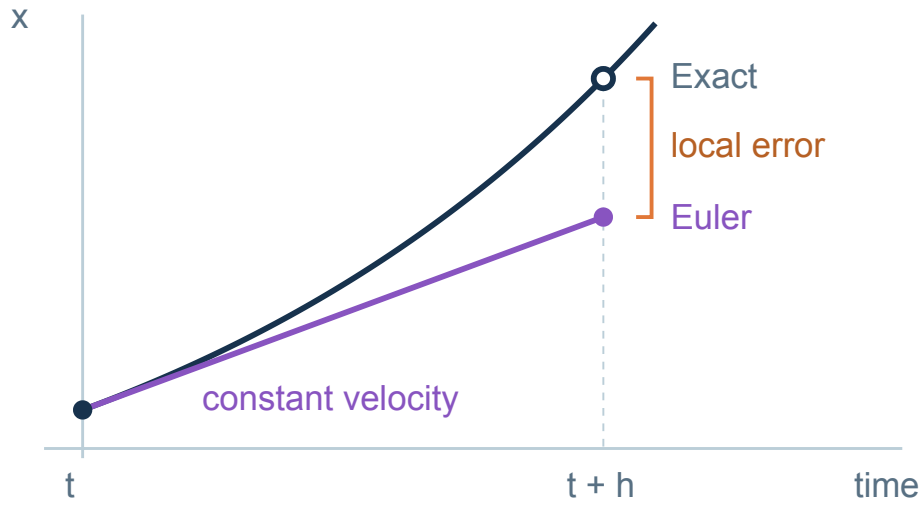
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Freeze the current velocity over the step: **one field evaluation**.

One step: follow the tangent



One-Step Error

For a smooth field, Taylor expansion from the **exact starting point** gives

$$\mathbf{x}(t+h) = \mathbf{x}(t) + h\,\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t) + \mathcal{O}(h^2).$$

Euler drops the higher-order terms: the **local error is** $\mathcal{O}(h^2)$.

Euler method; original one-step illustration
Driscoll T. A., Braun R. J. Fundamentals of Numerical Computation: Runge–Kutta methods

Numerical Solution of ODEs: Euler and Heun

Same initial point and step size: 8 steps on [0, 3], h = 0.375.

Euler

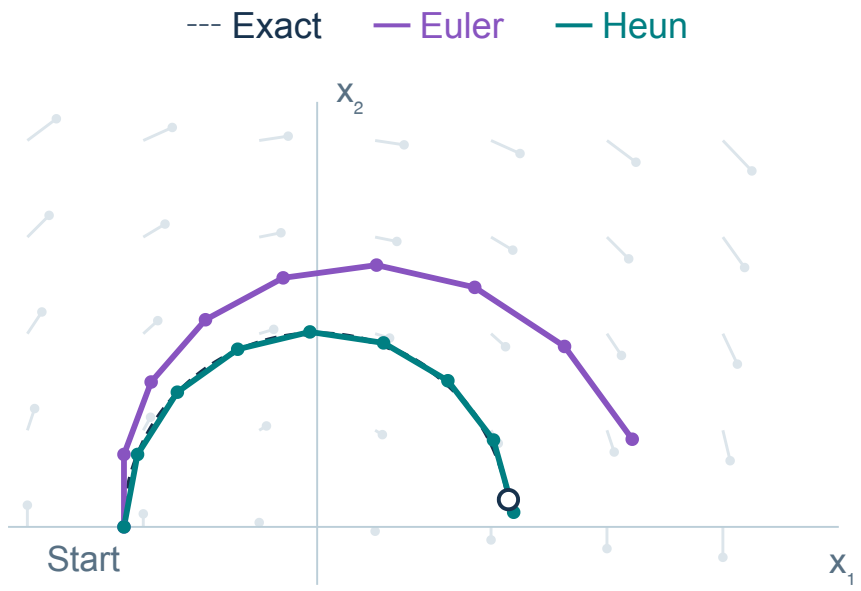
$$\mathbf{x}(t+h) = \mathbf{x}(t) + h\,\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

Heun: predict the endpoint

$$\mathbf{x}'(t+h) = \mathbf{x}(t) + h\,\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)$$

Heun: average the two velocities

$$\mathbf{x}(t+h) = \mathbf{x}(t) + \frac{h}{2}\big(\mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t) + \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}'(t+h), t+h)\big)$$



Error 0.713

Error 0.071

Step 8 / 8 · Field evaluations: Euler 8, Heun 16.

> Heun averages two velocities.
> Twice the field evaluations per step.

Example: $\mathbf{v}(\mathbf{x}) = (x_2, -x_1), \quad \mathbf{x}(0) = (-1, 0), \quad t \in [0, 3].$

ODE Solvers: Accuracy and Cost

For a sufficiently smooth velocity field on a fixed interval $[0, T]$:

$$\|\widehat{\mathbf{x}}_N - \mathbf{x}(T)\| = \mathcal{O}(h^p), \qquad h = T/N.$$

This is the **global error** after N steps.

Method	Global error	Velocity evaluations / step
Euler	$\mathcal{O}(h)$	1
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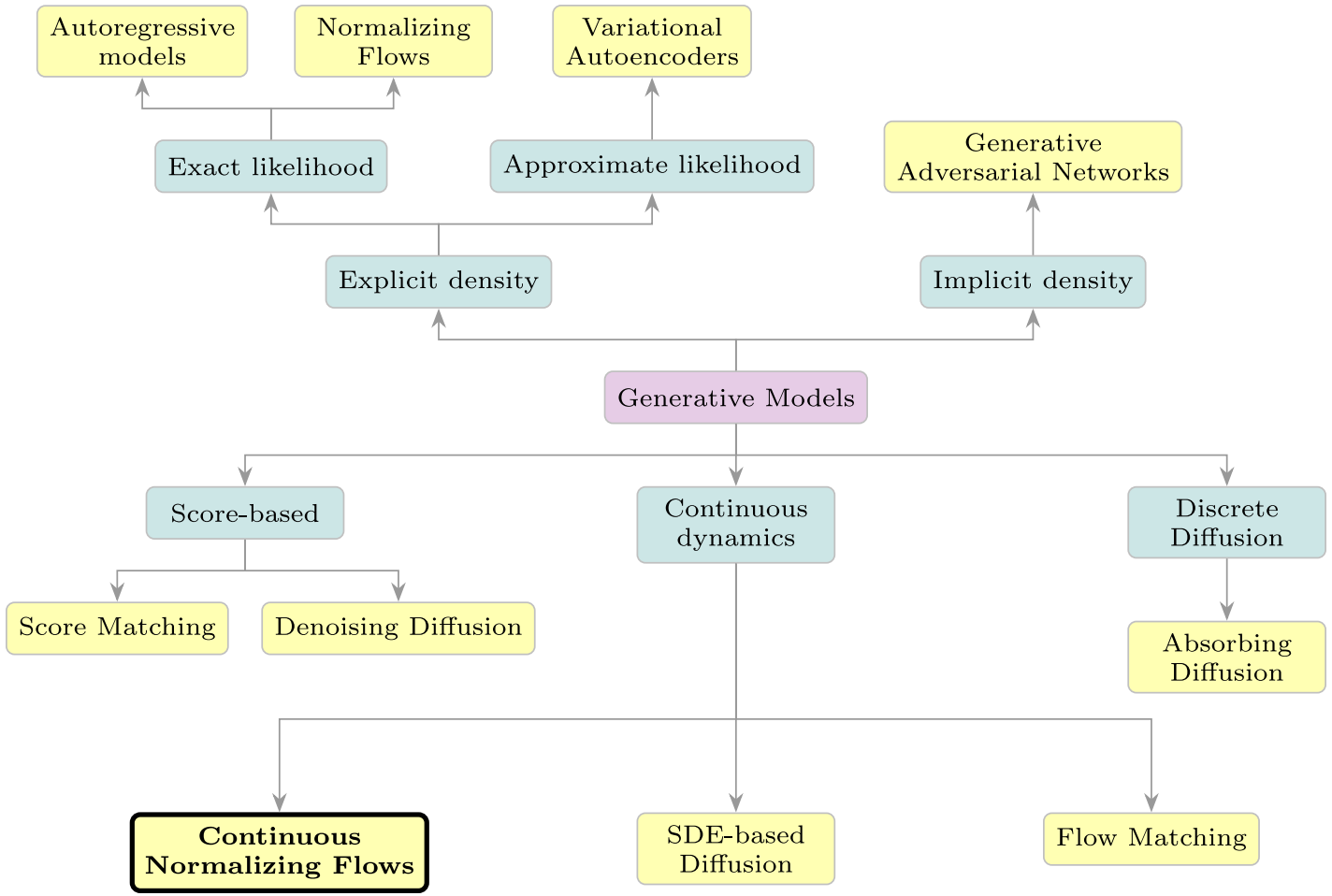
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- **One-step error** from an exact starting point is $\mathcal{O}(h^{p+1})$.
- For small h, halving the step roughly reduces global error by **2 / 4 / 16**, but doubles the number of steps.
- Compare computational cost by **NFE**: the number of velocity-network evaluations.

Generative Models Taxonomy



Continuous-Time Normalizing Flows: Neural ODE

Neural ODE

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

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- Consider $[t_0, t_1] = [0, 1]$ for simplicity.
- If $\mathbf{x}(0)$ is a random variable with density $p_0(\mathbf{x})$,
- Then, for any t, $\mathbf{x}(t)$ is a random variable with density $p_t(\mathbf{x})$.

Continuous-Time Normalizing Flows: Intuition

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

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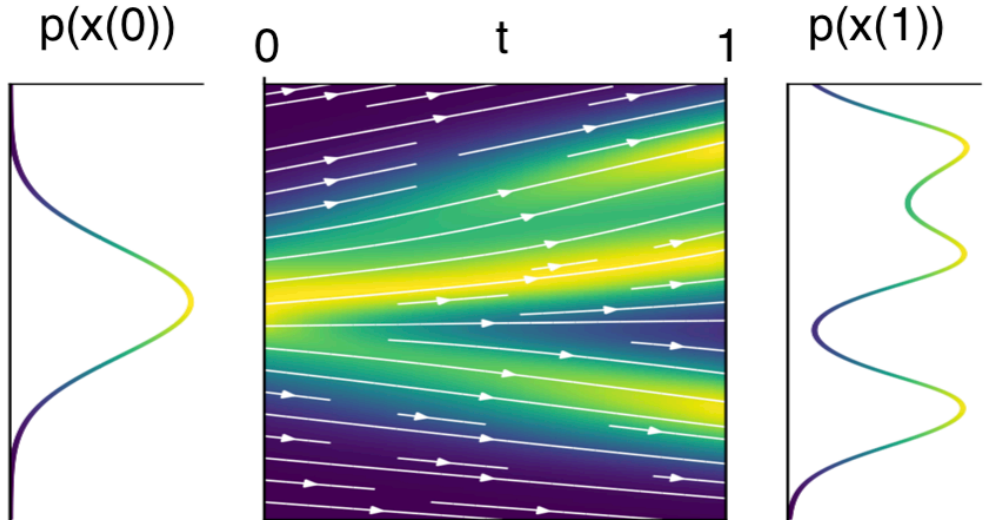
$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

- $p_t(\mathbf{x}) = p(\mathbf{x}, t)$ describes the **probability path** interpolating between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.
- What is the difference between $p_t(\mathbf{x}(t))$ and $p_t(\mathbf{x})$?

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Continuous-Time Normalizing Flows: Reversibility

Theorem (Picard)

If $\mathbf{v}$ is continuously differentiable with a bounded derivative in $\mathbf{x}$ and continuous in t, then the ODE has a **unique solution** given by a flow $\boldsymbol{\psi}_t$.

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If $\mathbf{v}$ is continuously differentiable with a bounded derivative in $\mathbf{x}$ and continuous in t, then the ODE has a **unique solution** given by a flow $\boldsymbol{\psi}_t$.

This guarantees the ODE is **uniquely reversible**.

$$\boldsymbol{\psi}_1(\mathbf{x}_0) = \mathbf{x}_0 + \int_0^1 \mathbf{v}_{\boldsymbol{\theta}}(\boldsymbol{\psi}_t(\mathbf{x}_0), t)\, dt$$

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How can we compute $p_t(\mathbf{x})$ at arbitrary t?

Outline

01 Continuous-Time Normalizing Flows (CNF)

02 Continuity Equation for CNF Log-Likelihood

03 SDE Basics

Continuous-Time NF

Theorem (Continuity Equation)

If $\mathbf{v}$ is continuously differentiable in $\mathbf{x}$ (with bounded derivative) and continuous in t, and $p_t(\mathbf{x}) > 0$, then

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018
Grathwohl W. et al. FFJORD: Free-form Continuous Dynamics for Scalable Reversible Generative Models, 2018

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Integrating gives the density **along the trajectory**:

Solution

$$\log p_1(\mathbf{x}(1)) = \log p_0(\mathbf{x}(0)) - \int_0^1 \operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) dt.$$

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Hutchinson's trace estimator:

$$\operatorname{tr}(\mathbf{J}) = \mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})}[\boldsymbol{\epsilon}^T \mathbf{J} \boldsymbol{\epsilon}], \qquad \mathbf{J} = \frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}.$$

- One noise vector gives an **unbiased trace estimate** via autodiff, without forming the full Jacobian.
- **Exact identity; approximate computation** using an ODE solver and a stochastic trace estimate.

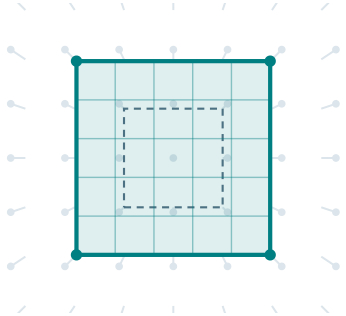
Chen R. T. Q. et al. Neural Ordinary Differential Equations, 2018
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Divergence: Where Does the Density Go?

Lecture 2: area change $\dfrac{A_t}{A_0} = |\det \mathbf{J}_{\psi_t}|, \qquad \mathbf{J}_{\psi_t} = \dfrac{\partial \psi_t}{\partial \mathbf{x}_0}$

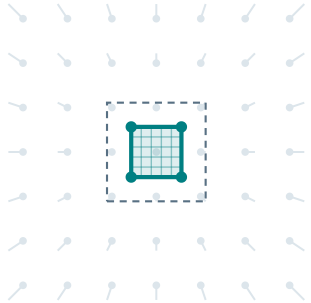
CNF: log-area rate $\dfrac{d}{dt} \log |\det \mathbf{J}_{\psi_t}| = \operatorname{div} \mathbf{v} = \operatorname{tr}\left(\dfrac{\partial \mathbf{v}}{\partial \mathbf{x}}\right)$

Expansion · $\mathbf{v} = 0.45\mathbf{x}$



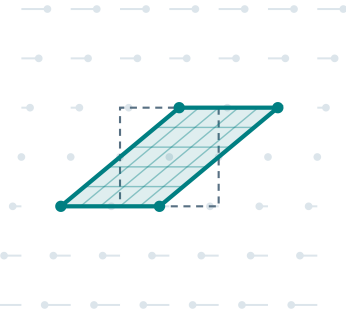
Divergence	**0.90**
Area	**3.857**
Density inside the patch	**0.259**
Area × density	**1.000**

Contraction · $\mathbf{v} = -0.45\mathbf{x}$



Divergence	**-0.90**
Area	**0.259**
Density inside the patch	**3.857**
Area × density	**1.000**

Shear · $\mathbf{v} = (0.8x_2, 0)$



Divergence	**0.00**
Area	**1.000**
Density inside the patch	**1.000**
Area × density	**1.000**

Mass is conserved: $\dfrac{d \log p_t(\mathbf{x}(t))}{dt} = -\dfrac{d \log A_t}{dt} = -\operatorname{div} \mathbf{v}.$

t = 1.5. Dashed: initial unit square; solid: moving patch. Area scaling is local in general; exact here (linear flows).

CNF: Training and Sampling

Fix $p_0 = \mathcal{N}(0, \mathbf{I})$ and learn $\mathbf{v}_{\boldsymbol{\theta}}$; the endpoint density is $p_1 = p_{\boldsymbol{\theta}}$.

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Training

1. Sample $\mathbf{x} \sim p_{\text{data}}$. Solve the ODE **backward**, from $\mathbf{x}(1) = \mathbf{x}$ to $\mathbf{x}(0)$.
2. Compute the negative log-likelihood along this trajectory:

$$\mathcal{L} = -\log p_{\boldsymbol{\theta}}(\mathbf{x}) = -\log p_0(\mathbf{x}(0)) + \int_0^1 \operatorname{tr}\left(\frac{\partial \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right) dt.$$

3. Update $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} \mathcal{L}$; differentiate through the ODE, e.g., via the **adjoint method**.

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Sampling

1. Sample $\mathbf{x}(0) \sim p_0$.
2. Solve the ODE **forward** and return the endpoint:

$$\mathbf{x}(1) = \texttt{ODESolve}_v(\mathbf{x}(0), \boldsymbol{\theta}, t_0 = 0, t_1 = 1) \sim p_{\boldsymbol{\theta}}.$$

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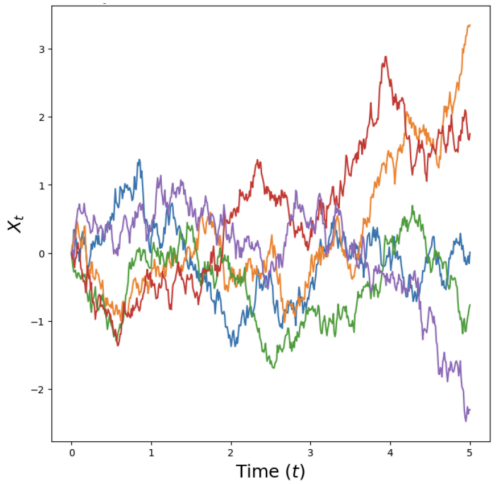
03 SDE Basics

Stochastic Differential Equation (SDE)

Wiener Process

$\mathbf{w}(t)$ is the standard Wiener process (Brownian motion), defined by:

1. $\mathbf{w}(0) = 0$ (almost surely);
2. $\mathbf{w}(t)$ has independent increments;
3. $\mathbf{w}(t)$ trajectories are continuous;
4. $\mathbf{w}(t) - \mathbf{w}(s) \sim \mathcal{N}(0, (t-s)\mathbf{I})$ for $t > s$;

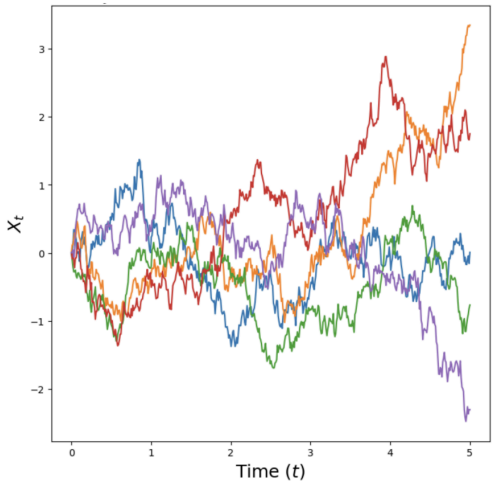


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$$d\mathbf{w} = \mathbf{w}(t + dt) - \mathbf{w}(t) = \mathcal{N}(0, \mathbf{I} \cdot dt) = \boldsymbol{\epsilon} \cdot \sqrt{dt}, \quad \text{where } \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

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Let's define a stochastic process $\mathbf{x}(t)$ with initial condition $\mathbf{x}(0) \sim p_0(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}$$

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- $\mathbf{f}(\mathbf{x}, t) : \mathbb{R}^m \times [0, 1] \to \mathbb{R}^m$ is the **drift** term (vector field).
- $g(t) : \mathbb{R} \to \mathbb{R}$ is the **diffusion** term (if $g(t) = 0$, we recover the standard ODE).
- $\mathbf{w}(t)$ is the standard Wiener process ($d\mathbf{w} = \boldsymbol{\epsilon} \cdot \sqrt{dt}$).
- We do not have the flow $\boldsymbol{\psi}_t(\mathbf{x}_0)$ notion anymore, since trajectories are stochastic.

Itô Integration Basics

The SDE should be understood in its **integral form** (Itô sense):

$$\mathbf{x}(t) = \mathbf{x}(0) + \int_0^t \mathbf{f}(\mathbf{x}(s), s)\, ds + \int_0^t g(s)\, d\mathbf{w}(s)$$

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$$\int_0^t g(s)\, d\mathbf{w}(s) = \lim_{N\to\infty} \sum_{i=0}^{N-1} g(t_i)\big(\mathbf{w}(t_{i+1}) - \mathbf{w}(t_i)\big)$$

- Brownian paths are continuous but a.s. **nowhere differentiable** $\Rightarrow$ Fundamental Theorem of Calculus fails.
- **Itô's lemma** is the stochastic counterpart of the chain rule; we will not need its explicit form here.
- Expressions $d\mathbf{x}$, dt, $d\mathbf{w}$ are **formal shorthand** for infinitesimal increments.

Stochastic Differential Equation (SDE)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}$$

Theorem

If $\mathbf{f}$ is continuously differentiable with a bounded derivative in $\mathbf{x}$ and continuous in t and $g(t)$ is continuous then the SDE has the solution given by unique process $\mathbf{x}(t)$.

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- Unlike ODEs, the initial condition $\mathbf{x}(0)$ doesn't uniquely determine the trajectory.
- There are two sources of randomness:
  - the initial distribution $p_0(\mathbf{x})$;
  - the Wiener process $\mathbf{w}(t)$.

Stochastic Differential Equation (SDE)

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Discretizing the SDE (Euler-Maruyama Update) – `SDESolve`

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}(t), t) \cdot dt + g(t) \cdot \boldsymbol{\epsilon} \cdot \sqrt{dt}$$

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- At any time t, the process has density $p_t(\mathbf{x}) = p(\mathbf{x}, t)$.
- $p : \mathbb{R}^m \times [0, 1] \to \mathbb{R}_+$ specifies a **probability path** from $p_0(\mathbf{x})$ to $p_1(\mathbf{x})$.
- How can we obtain the probability path $p_t(\mathbf{x})$ for $\mathbf{x}(t)$?

Stochastic Differential Equation (SDE)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}, \quad d\mathbf{w} = \boldsymbol{\epsilon} \cdot \sqrt{dt}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).$$

Theorem (Kolmogorov-Fokker-Planck)

If $p_t(\mathbf{x}) \in C^{1,2}$ (i.e., C^1 in t and C^2 in $\mathbf{x}$), then

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}\left(\mathbf{f}(\mathbf{x}, t) p_t(\mathbf{x})\right) + \frac{1}{2} g^2(t) \Delta_{\mathbf{x}} p_t(\mathbf{x})$$

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$$\operatorname{div}(\mathbf{v}) = \sum_{i=1}^{m} \frac{\partial v_i(\mathbf{x})}{\partial x_i} = \operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x})}{\partial \mathbf{x}}\right)$$

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Stochastic Differential Equation (SDE)

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Let $g(t) = 1$, then $\mathbf{f}(\mathbf{x}, t) = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$.

$$d\mathbf{x} = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})\, dt + 1 \cdot d\mathbf{w}$$

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Discretized Langevin SDE

$$\mathbf{x}_{t+1} - \mathbf{x}_t = \frac{\eta}{2} \cdot \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}, \quad \eta \approx dt.$$

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Setting the stationary density to the model distribution, $p_t(\mathbf{x}) = p_{\boldsymbol{\theta}}(\mathbf{x})$ for all t:

Langevin Dynamic

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}, \quad \eta \approx dt.$$

We (partially) explained, why Langevin dynamics is working.

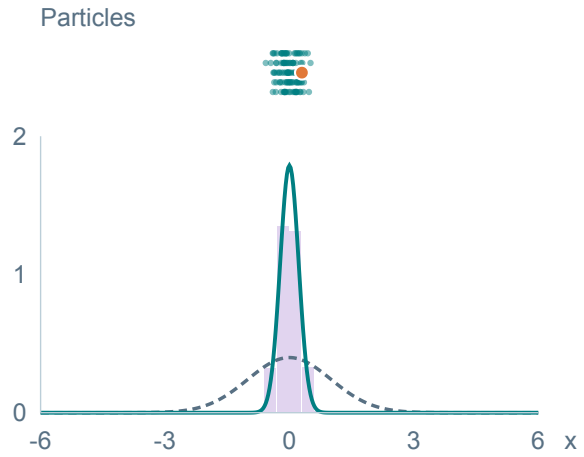
Langevin: Moving Particles, Stationary Density

$$p_0(x) = p_*(x) = \mathcal{N}(0, 1), \qquad \tfrac{1}{2}\nabla_x \log p_*(x) = -\tfrac{1}{2}x$$

Same initial samples from the target. All three cases at t = 3.

▥ Empirical histogram — Exact current density --- Target N(0, 1)

Drift only · $dx = -\frac{1}{2}x\,dt$



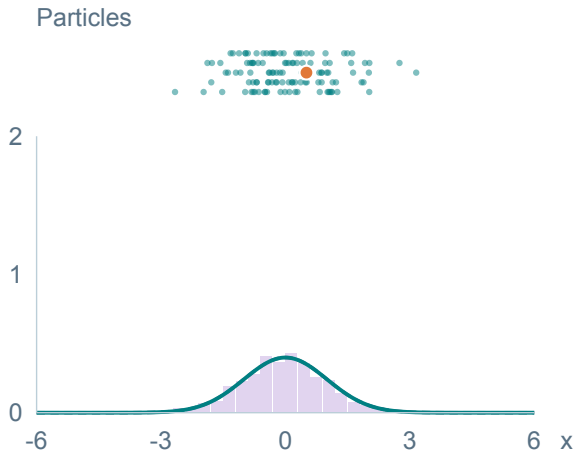
Exact variance: 0.050

Diffusion only · $dx = dw$



Exact variance: 4.000

Both · $dx = -\frac{1}{2}x\,dt + dw$



Exact variance: 1.000

> Stationary density does not mean stationary particles.

768 samples; finite-sample fluctuations remain. Exact Gaussian transitions isolate stationarity from discretization error.

Summary

- CNFs define invertible transformations via neural ODEs; likelihood-based training requires ODE solves, trace evaluations and gradients through the solver.

- If $\mathbf{x}_0$ is a random variable, this yields a **probability path** $p_t(\mathbf{x})$ as time evolves. The continuity equation describes the evolution of $\log p(\mathbf{x}, t)$ over time.

- An SDE defines a stochastic process with drift and diffusion terms; ODEs are a special case of SDEs.

- The KFP equation describes the probability dynamics of an SDE.

- The Langevin SDE preserves a constant probability path.